

Global Trade in Low Carbon Technology Products

1994-2023

Objective :

This report analyzes global trade in low-carbon technology products across 210 countries from 1994-2023. It highlights trade trends ,leading countries ,product categories and business insights.

Dataset Information

	Values
Records	48,367
Countries	210
Indicators	11
Product categories	5
Time Period	1994-2023

- Cover 210 countries.
- Includes import ,export ,total trade and trade balance.
- Contains 11 economic indicators.
- cover 30 years(1994-2023).

210

Count of Country

11

Count of Indicator

5

Count of CTS_Name

30

Count of Year

Global Trade Performance

Business Question :

How has global trade in low-carbon technology products evolved from 1994 to 2023?

Year

Country

All

All

Total

66.53T

Highest

2.24T

Lowest

25.04K

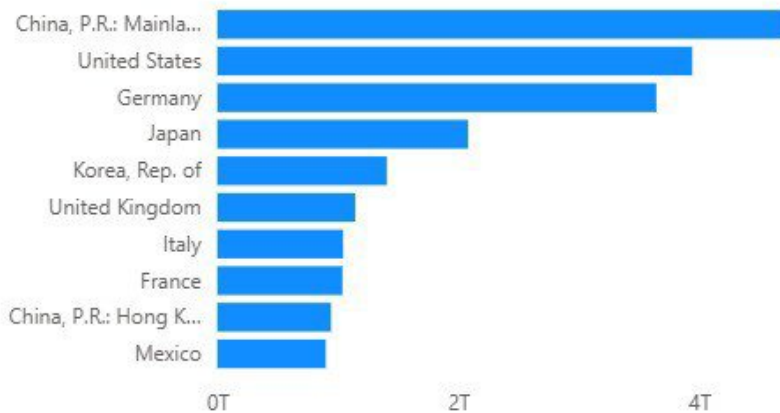
Average

14.50bn

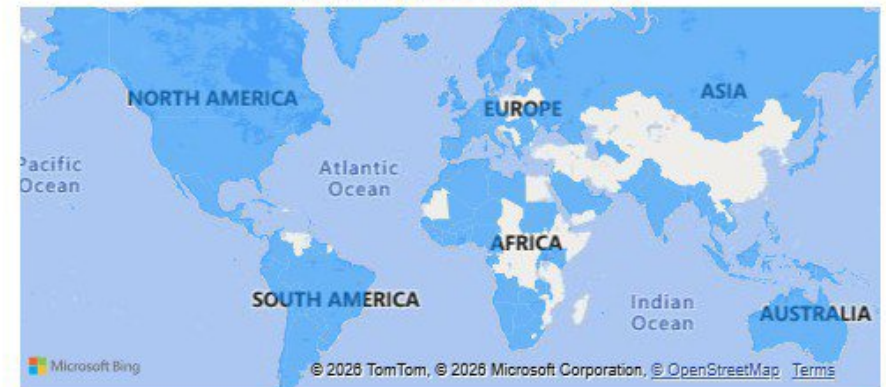
Global Trade Trend



Top 10 Countries



Global Trade Distribution



Business Insights

- Trend increased over time
- China leads global trade
- US & Germany follow
- Trade concentrated in few nations

Trade Dependency Analysis

Business Question :

How do imports and exports compare across countries, and which economies are more dependent on global low-carbon technology trade?

Year

All

Country

All

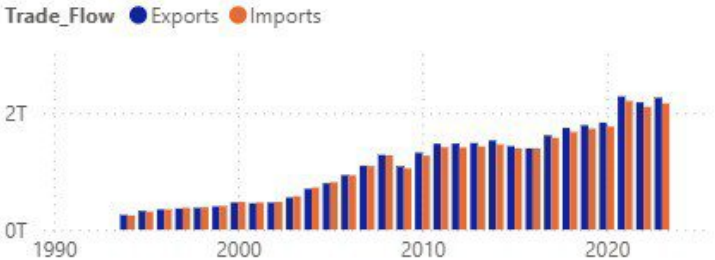
Total Export

33.67T

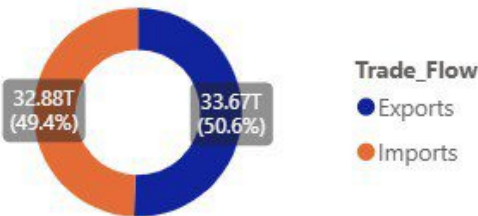
Total Import

32.88T

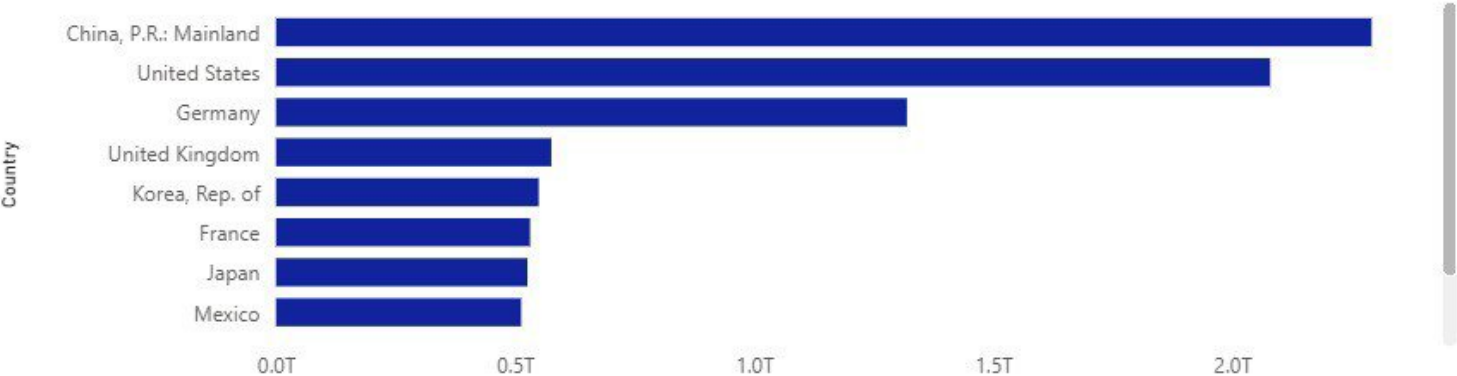
Import vs Export Trend



Import vs Export Share



Top 10 Importing Countries



Business Insights

- Export trade (33.67T) is slightly higher than import trade (32.88T), reflecting strong global export activity.
- Import and export values have increased consistently over the last three decades, showing continuous growth in the low-carbon technology market.
- A few countries dominate international trade, whereas the majority of countries have comparatively lower trade volumes.
- The balanced trade distribution indicates a well-connected global supply chain for low-carbon technology products.

Trade Competitiveness Analysis

Business Question :

Which countries show the strongest trade performance relative to their economy in low-carbon technology products?

Year

All

Country

All

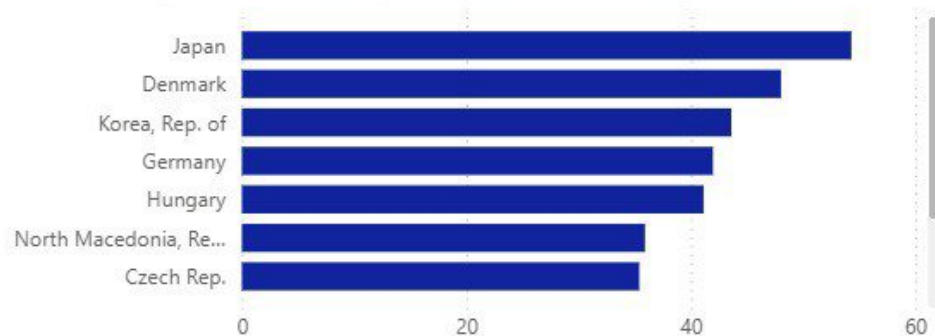
Highest Comparative
Advantage Index

7.74

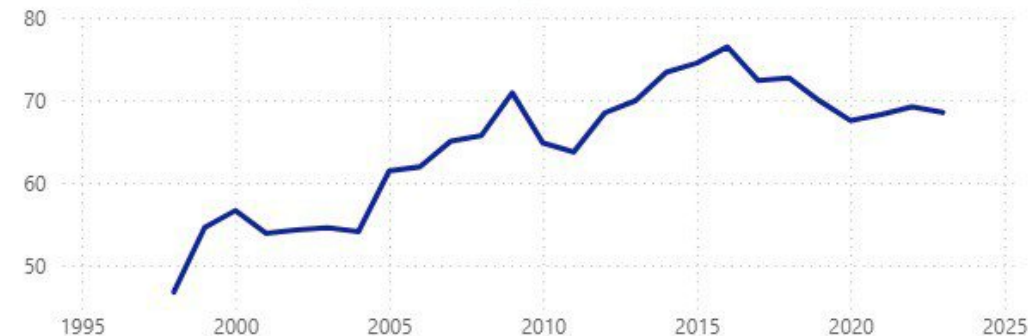
Average Comparative
Advantage Index

0.42

Top 10 Countries by Comparative Advantage Index



Comparative Advantage Trend (1994–2023)



Business Insights

- Some countries consistently maintain a stronger comparative advantage in low-carbon technology products.
- Comparative advantage varies across countries, reflecting differences in technological capability and export specialization.
- Several economies show improving competitiveness over time.
- Comparative advantage helps identify countries with stronger global positioning in sustainable technology markets.

Economic Impact Analysis

Business Question :

How significant is low-carbon technology trade in national economies based on its contribution to GDP?

Year

All

Country

All

Country	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	Total
China, P.R.: Hong Kong	16.27	14.26	13.57	14.88	15.93	14.75	15.95	17.77	11.02	8.93	143.34
Czech Rep.	7.24	7.27	7.30	7.54	7.71	8.05	9.63	12.04	12.72	13.81	93.31
Hungary	8.77	9.28	10.13	9.70	9.29	10.20	11.94	13.76	17.13	17.66	117.85
Singapore	10.67	10.87	10.71	10.48	10.04	10.29	10.84	10.29	8.01	7.25	99.45
Slovak Rep.	7.06	7.48	7.74	6.95	6.54	9.32	13.84	17.47	19.66	19.73	115.80
Total	50.01	49.17	49.45	49.54	49.51	52.60	62.20	71.33	68.55	67.38	569.74

Countries Above Average GDP Contribution

115

Average GDP Contribution (%)

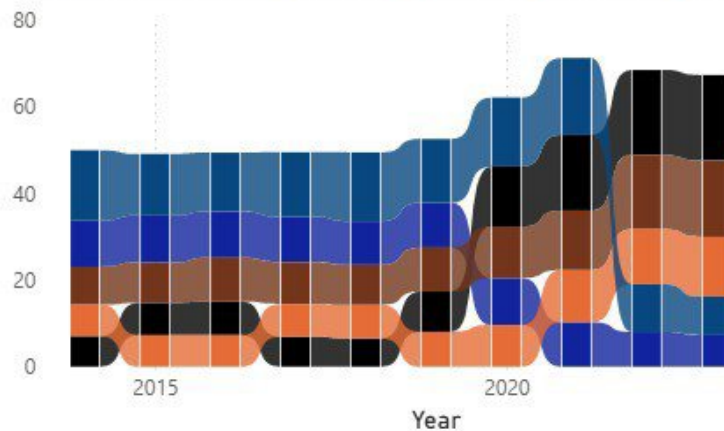
1.76

Highest GDP Contribution (%)

19.73

Trade Contribution to GDP Over Time

Country ● Slovak Rep. ● Singapore ● Hungary ● Czech Rep. ● China, P.R.,...



Business Insights

- Low-carbon technology trade contributes an increasing share to GDP across many economies.
- Countries such as China (Hong Kong), Hungary and the Slovak Republic record relatively higher GDP contributions in recent years.
- The average GDP contribution remains much lower than the highest observed value, indicating significant differences among countries.
- GDP dependence on low-carbon technology trade has strengthened steadily since the early 2000s.

Future Trade Outlook

Business Question :

Which countries have consistently led global low-carbon technology trade, and what strategic insights can guide future decisions?

Year

All

Country

All

Total Trade Value

66.53T

Countries with Active Trade

204

Highest Trade Value

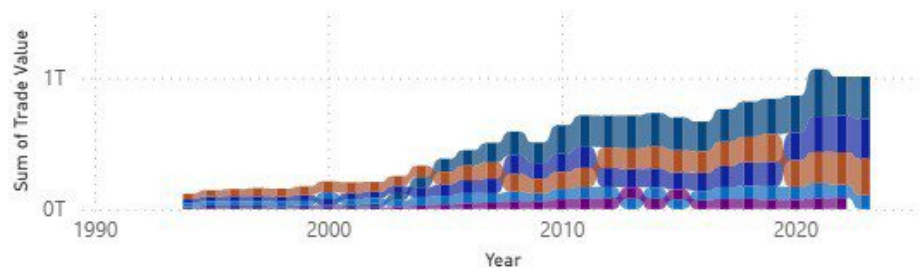
2.24T

Average trade Value

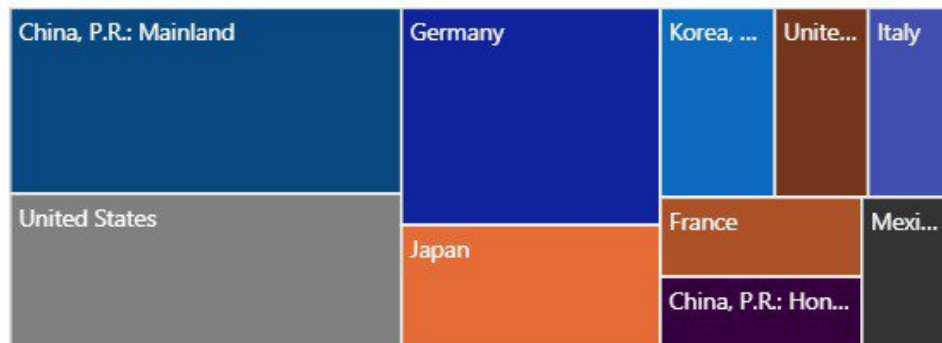
14.50bn

Global Trade Leadership Over Time

Country ● China, P.R.: Mainland ● Germany ● Japan ● Korea, Rep. of ● United States



Global Trade Distribution



Business Insights

- Global low-carbon technology trade has expanded steadily since 1994.
- Trade leadership remains concentrated among a few major economies.
- Historical trends indicate continued opportunities for sustainable trade growth.

Recommendations

- Prioritize partnerships with countries showing long-term trade leadership.
- Diversify supply chains to reduce dependence on a small number of markets.
- Monitor emerging economies with accelerating low-carbon trade growth